

Erik An

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profile

Software engineering background from the 42 School; now studying mathematics, aiming at computational neuroscience and brain-computer interfaces. Conventions I hold to: one canonical layout per project type, names that survive grep, math documented beside the code that implements it.

education

University of Vienna · 2025–present

BSc Mathematical Foundations of Data Science.

42 Vienna, Advanced · 2025–present

Data science specialization.

42 Wolfsburg, Core Curriculum · 2024–2025

Software engineering. Completed in 10 months.

British curriculum schools, Spain and UK · 2019–2024

A-Levels in Mathematics, Physics, and History, completed at Xabia International College.

projects

mfds · executable knowledge base

Study repository for my BSc, kept in plain text and run as a software project: Org-mode, LaTeX, Python and Julia, with the directory layout specified in writing and enforced by a pre-commit validator.

total_perspective_vortex · EEG motor-imagery classification

Common Spatial Patterns implemented from scratch as a scikit-learn transformer, feeding LDA; 0.63 mean held-out accuracy (6 binary task groups) across 109 subjects of PhysioNet's EEGMMIDB, with filters checked against mne.decoding.CSP.

audio_print · audio fingerprinting in Julia

Shazam-style matching through spectral peaks and combinatorial hashing over a SQLite database; identifies 97.8% of 138 noisy clips against 8,000 songs at ~260 ms per query.

transcendence · full-stack web app, team of three

Browser Pong platform with accounts, 2FA, tournaments and three microservices; my part was the game engine and AI opponent, most of the TypeScript frontend (no framework), and half of the match service.

webserv · HTTP/1.1 server in C++, team of three

Non-blocking C++98 server on a single epoll event loop, with an incremental request parser, CGI, file uploads and NGINX-style configuration.

ft_linear_regression · linear regression in Julia

Univariate model from scratch, trained by gradient descent with feature normalization; Makefile-driven train, predict and evaluate pipeline.

conferences

10th Graz Brain-Computer Interface Conference · attendee, September 2026

skills

Languages: Python, C/C++, Julia, TypeScript, JavaScript, SQL.

Libraries: NumPy, scikit-learn, MNE.

Tools: Docker, Git, Emacs, LaTeX.